# Chenru Wen

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## Research Interests

I work on robot learning for manipulation, focusing on recovery from execution failures by forecasting the outcomes of candidate actions. I am currently exploring how physical estimates and their uncertainty should enter dexterous policies.

## Education

**Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)** Aug 2025 – present
*M.Sc. in Robotics. Advisors: Xingxing Zuo, Dezhen Song* Abu Dhabi, UAE

- **Research Intern** at OpenDriveLab, Shanghai Innovation Institute, Jun 2026 – Sep 2026

**Tsinghua University, Yao Class (IIIS)** Sep 2021 – Jun 2025
*B.Eng. in Computer Science and Technology* Beijing, China

- **Visiting Research Intern** at UC San Diego, Feb 2024 – Aug 2024

## Publications

**ForkCast: Predictive Recovery for Vision-Language-Action Policies via Counterfactual Futures**

**Chenru Wen***, Huajian Zeng*, Haokai Ding, Jian Kang, Anqing Duan, Dawei Zhou, Dezhen Song, Xingxing Zuo

Under review at IEEE International Conference on Robotics and Automation (ICRA), 2027.

arxiv.org/abs/2610.18650

**Lay-Your-Scene: Natural Scene Layout Generation with Diffusion Transformers**

Divyansh Srivastava, Xiang Zhang, He Wen, **Chenru Wen**, Zhuowen Tu

In IEEE/CVF International Conference on Computer Vision (ICCV), 2025.

arxiv.org/abs/2505.04718

## Research Experience

**Predictive recovery for frozen VLA policies (ForkCast)** Oct 2025 – Sep 2026
*RCL, MBZUAI. Advisor: Xingxing Zuo* ICRA 2027, under review

- **Role.** Led all technical work: problem formulation, method, branched data, selector training, and the simulation and real-robot experiments.
- **Method.** Framed recovery as choosing among continuing the policy's chunk, moving the camera and re-querying, or replanning the arm, ranked by a candidate-conditioned latent world model and a shared progress head.
- **Simulation.** On five RoboTwin 2.0 tasks with injected occlusion and slip, frozen $\pi_{0.5}$ went from 13.3% to 30.0% pooled success (best fixed repair: 21.7%). The selector, trained only on $\pi_{0.5}$ branches, carried over to RDT and XR-0 without retraining.
- **Real robot.** On the platform below, fine-tuned $\pi_{0.5}$, collected real branches, and ran 30 trials per method in four settings (clean and restricted-visibility sponge transfer and can insertion); success rose in all four.

**Localization on non-metric maps** Oct 2025 – present
*MBZUAI. Advisor: Dezhen Song* In preparation

- Localize a walking camera on a printed visitor map, with no GPS or metric map, from local odometry and a few named landmarks: enumerate map alignments consistent with landmark bearings, then read out the nearest facility at each query.
- Evaluated on full walking sequences in Oxford; found that map-side feasibility constraints are necessary but carry little information, and that the reference itself caps attainable accuracy.

**Causal analysis of appearance shifts in VLA policies** Jun 2026 – Sep 2026
*Research Intern, OpenDriveLab, Shanghai Innovation Institute* Shanghai, China

- Turned a real-robot case, where changing only a target's appearance changed the grasp, into matched-state experiments with a fine-tuned $\pi_{0.5}$ on RoboTwin 2.0.
- Used probing and activation replacement across the visual encoder, VLM and action expert. Replacing full visual or camera-level representations recovered the donor's actions; finer interventions failed the pre-set stability bars.

## Engineering

**Real-robot VLA platform (dual-arm AgileX PiPER)** Dec 2025 – Sep 2026
*RCL, MBZUAI. Used for $\pi_0$ / $\pi_{0.5}$ deployment and all ForkCast real-robot experiments* Abu Dhabi, UAE

- **Hardware.** Two PiPER arms; a RealSense head camera on an IQR PTU2 pan/tilt unit mounted on a lift that also carries both arm bases; two wrist cameras; a U-Arm leader arm for teleoperation.
- **System engineering.** Adapted openpi to the robot: a ROS 2 inference client, a Docker policy server, and real-time action chunking. Wrote the pan/tilt and lift control, a library of camera views, and the ForkCast selector service.
- **Debugging.** Recalibrated the lift's absolute height against its lower limit switch after the encoder silently re-zeroed; traced a grasp failure to a gripper command clamped at 80 mm while demonstrations opened to 100 mm.
- **Data and evaluation.** Collected demonstrations by teleoperation and replay with one-command collection and inference scripts, and ran every real-robot trial reported in ForkCast.

## Honors and Awards

**Lenovo Scholarship**, Tsinghua University 2024
**First-Class Freshman Scholarship**, Tsinghua University 2021

## Service

Reviewer: IEEE International Conference on Robotics and Automation (ICRA) 2027.

## Skills

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| **Programming** | Python, C/C++, Bash |
| **Learning** | PyTorch; RL for dexterous hands; VLA fine-tuning ($\pi_0$, $\pi_{0.5}$); latent world models |
| **Simulation** | Isaac Lab, ManipTrans / OakInk2, RoboTwin 2.0 (SAPIEN) |
| **Robots** | AgileX PiPER dual-arm, pan/tilt and lift camera rigs, Intel RealSense |